SozKZ: Training Efficient Small Language Models for Kazakh from Scratch

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Kazakh, spoken by 22 million people, lacks dedicated language models. Existing multilingual LLMs allocate minimal capacity to low-resource languages and suffer from tokenizer inefficiency on agglutinative morphology. We present SozKZ, a family of Llama-architecture models (50M–600M parameters) trained from scratch on 9 billion tokens of Kazakh text with a purpose-built 32K BPE tokenizer. On held-out Kazakh text, our 600M model achieves 0.847 bits per byte, outperforming all evaluated multilingual models including those with over 100× more parameters. On sentiment classification, SozKZ-600M reaches 65.4%, competitive with 8B-parameter multilingual models. We observe clear limitations on knowledge-intensive tasks: multiple-choice QA (15.6%) and reading comprehension (33.4%) remain below larger models that benefit from extensive world knowledge encoded during training on much larger corpora. These results demonstrate that small dedicated models offer a compelling efficiency–performance trade-off for language-centric tasks, while knowledge-intensive benchmarks still require larger-scale training. We release all models, tokenizer, and training pipeline under open licenses.

1 Introduction

Kazakh is a Turkic language spoken by approximately 22 million people, primarily in Kazakhstan and neighboring countries. As an agglutinative language with rich morphology, Kazakh poses particular challenges for natural language processing: words carry extensive inflectional and derivational suffixes, leading to a large effective vocabulary that general-purpose multilingual tokenizers handle poorly. Despite a growing body of NLP research for Kazakh [25, 26], the language remains underserved by the current generation of large language models, which typically allocate only a fraction of their capacity to low-resource languages.

Modern multilingual models such as Llama 3 [23], Gemma [9], Qwen 2.5 [4], and Mistral [11] are trained on corpora dominated by English and other high-resource languages. When applied to Kazakh, these models face two compounding inefficiencies. First, their tokenizers fragment Kazakh text into disproportionately many subword tokens—a phenomenon quantified by *fertility*, the average number of tokens per word. Second, the vast majority of model parameters encode knowledge about languages other than Kazakh, yielding a poor ratio of useful capacity to total model size.

We argue that **small, dedicated models can match or exceed much larger multilingual models on Kazakh language tasks with orders-of-magnitude better efficiency**. To test this thesis, we train SozKZ, a family of Llama-architecture causal language models at four parameter scales—50M, 150M, 300M, and 600M—entirely from scratch on a curated Kazakh corpus of approximately 9 billion tokens.

Our contributions are as follows:

1. We present an end-to-end pipeline for training Kazakh language models from scratch at four scales (50M–600M parameters), including data collection, cleaning, and pre-tokenization.
2. We design a dedicated ByteLevel BPE tokenizer with a 50K vocabulary[1] that achieves a 2–3× fertility advantage over multilingual tokenizers on Kazakh text.
3. We construct a 6-task benchmark suite and evaluate SozKZ models against 14 competitors spanning 0.5B–120B parameters, providing the most comprehensive comparison of language model performance on Kazakh to date.
4. We report empirical scaling curves for Kazakh language modeling, identifying the efficiency sweet spot where dedicated small models offer the best performance-per-parameter trade-off.
5. We release all models, the tokenizer, and the training pipeline under open licenses to support further research on Kazakh NLP.

The rest of this paper is organized as follows. Section 2 surveys related work on Kazakh NLP, low-resource language modeling, scaling laws, and tokenizer design. Section 3 describes our data pipeline, tokenizer, model architectures, and training procedure. Section 4 details the benchmark suite and evaluation protocol. Section 5 presents and analyzes results. Section 6 summarizes our findings and outlines future directions.

2 Related Work

2.1 Kazakh NLP

Research on Kazakh natural language processing has accelerated in recent years, driven largely by the IS2AI research group at Nazarbayev University. Yeshpanov et al. [25] introduced KazNERD, a large-scale named entity recognition dataset for Kazakh covering 25 entity classes across news, fiction, and legal domains. Yeshpanov et al. [26] subsequently released KazQAD, a question-answering dataset for Kazakh built from Wikipedia and educational texts. These resources have enabled systematic evaluation of NLP models on Kazakh for the first time.

Beyond datasets, several efforts have targeted Kazakh language modeling directly. Multilingual models such as mBERT [8] and XLM-R [7] include Kazakh in their training data but allocate minimal capacity to it. More recently, KazLLM [28] explored continued pre-training of large models on Kazakh corpora. Our work differs in training dedicated models entirely from scratch, allowing full control over architecture, tokenizer, and data composition.

2.2 Low-Resource Language Models

Training language models for low-resource languages has emerged as an active research area. Ogueji et al. [16] demonstrated with AfriBERTa that small transformer models trained on modest corpora (less than 1 GB) can achieve competitive performance on African languages, challenging the assumption that large-scale data is a prerequisite. For Arabic, Sengupta et al. [18] trained Jais, a 13B-parameter model from scratch on a bilingual Arabic-English corpus, showing that dedicated models outperform multilingual ones on Arabic-specific benchmarks.

SEA-LION [3] adopted a similar philosophy for Southeast Asian languages, training models at multiple scales with language-specific tokenizers. The BLOOM project [6] took a different

[1] `https://huggingface.co/stukenov/kazakh-bpe-32k`

approach, training a single 176B-parameter multilingual model on 46 languages with the goal of equitable coverage. Our work follows the dedicated-model approach of Jais and AfriBERTa but at substantially smaller scales (50M–600M parameters), testing how far efficiency gains from specialization can compensate for reduced model capacity.

2.3 Scaling Laws

Kaplan et al. [12] established power-law relationships between model size, dataset size, compute budget, and cross-entropy loss for neural language models. Hoffmann et al. [10] refined these findings with the Chinchilla scaling laws, demonstrating that many large models are significantly undertrained relative to their parameter count and that a compute-optimal model should be trained on approximately 20 tokens per parameter.

In practice, the Llama model family [23, 24] demonstrated that training smaller models on substantially more data than the Chinchilla-optimal ratio yields models that are more efficient at inference time. Our training regime follows this over-training philosophy: the SozKZ-600M model is trained on approximately 9B tokens, yielding a tokens-to-parameters ratio of roughly 15:1—slightly below the Chinchilla optimum but reflective of the available Kazakh data.

These scaling laws have been derived primarily from English-language experiments. Whether they transfer to morphologically rich, low-resource languages with different tokenizer efficiencies remains an open question that our scaling analysis addresses.

2.4 Tokenizer Design

Subword tokenization, introduced by Sennrich et al. [19] with Byte-Pair Encoding (BPE) and refined by Kudo and Richardson [13] with SentencePiece, is now standard in language model pre-training. The choice of tokenizer has outsized effects on model efficiency for morphologically rich languages [17]: a tokenizer trained on English-dominated data will fragment agglutinative languages like Kazakh, Turkish, or Finnish into far more tokens per word than necessary, effectively reducing the model's context window and increasing inference cost per semantic unit.

Fertility—the average number of tokens per whitespace-separated word—is a widely used metric for quantifying this inefficiency [1]. Prior work on Turkic languages has shown that dedicated tokenizers achieve fertility values 2–3× lower than multilingual alternatives [22].

We train a ByteLevel BPE tokenizer with a 50K vocabulary exclusively on Kazakh text and quantify its fertility advantage over the tokenizers used by Llama 3, Gemma, Qwen 2.5, and Mistral. This dedicated tokenizer is a key component of the efficiency gains we observe, as lower fertility translates directly to more content per fixed context window.

3 Methodology

This section describes the full training pipeline for the SozKZ model family: data collection and cleaning, tokenizer design, model architecture, and training configuration. All code, configs, and trained models are released publicly to enable full reproducibility.

3.1 Training Data

We construct a large-scale monolingual Kazakh corpus by aggregating text from 18 web and curated sources, including CulturaX [7], HPLT 2.0, mC4 [8], MADLAD-400, mOSCAR, Kazakh Wikipedia, and the `kz-transformers/multidomain-kazakh-dataset`. The raw collection comprises 28.4M documents.

We apply a 9-stage cleaning pipeline:

1. Unicode NFC normalization,
2. removal of control characters,
3. whitespace and newline collapsing,
4. minimum length filtering (≥ 50 characters),
5. URL density filtering (≤ 5 per 1,000 characters),
6. HTML tag filtering (≤ 5 tags),
7. Kazakh character ratio filter (script-based),
8. language identification (retaining only Kazakh),
9. exact deduplication via MD5 hashing (both within-source and cross-source against the existing `multidomain-kazakh-dataset` reference of 12.4M unique hashes).

After cleaning, 13.7M documents remain (48.2% pass rate), yielding approximately 9.0B tokens under our BPE 50K tokenizer. This constitutes one of the largest publicly available monolingual Kazakh corpora. We note the inherent data-quality versus quantity tradeoff in low-resource settings: aggressive filtering removes noise but discards potentially useful text, while lenient filtering risks training on low-quality data. Our 48.2% pass rate reflects a moderately conservative stance.

3.2 Tokenizer

We train a ByteLevel BPE tokenizer [19] with a vocabulary of 50,257 tokens exclusively on Kazakh text from our cleaned corpus.

The motivation for a dedicated tokenizer is twofold. First, multilingual tokenizers (e.g., those used by Llama, Qwen, or Gemma) allocate vocabulary capacity across 100+ languages, resulting in poor *fertility*—the average number of tokens per word—for morphologically rich, Cyrillic-script languages like Kazakh [17, 22]. Second, better tokenization directly improves training and inference efficiency: fewer tokens per document means faster throughput and lower compute cost for a fixed amount of text.

The tokenizer is published as open-source on HuggingFace (`saken-tukenov/sozkz-core-gpt2-50k-kk-base-v`

3.3 Model Architecture

We train four model sizes following the LlamaForCausalLM architecture [23]: SwiGLU activations [20], Rotary Position Embeddings (RoPE) [21], RMSNorm pre-normalization [27], and tied input–output embeddings. All models are trained from scratch with no pretrained initialization.

The Llama architecture was chosen for its well-understood scaling properties [12], training stability, and broad ecosystem support. The intermediate dimension follows a $3.5\times$ multiplier of the hidden size across all configurations, consistent with SwiGLU best practices [20]. Context length is 2,048 tokens for the 50M, 300M, and 600M models; the 150M model uses a 1,024-token context.

1: Architecture configurations for the SozKZ model family. All models use SwiGLU, RoPE, RMSNorm, tied embeddings, and a vocabulary of 50,257 tokens.

Model	Params	Layers	Hidden	Heads	Intermediate
SozKZ-50M	50.3M	8	576	8	1,536
SozKZ-150M	151.9M	16	768	12	2,048
SozKZ-300M	325M	18	1,024	16	3,584
SozKZ-600M	587M	22	1,280	20	4,480

3.4 Training Details

All models are trained for one epoch over the full ∼9.0B-token corpus using the AdamW optimizer [15] with $\beta_1 = 0.9$, $\beta_2 = 0.95$, and weight decay of 0.1. We use a cosine learning rate schedule [14] with linear warmup. Peak learning rates are 6×10^{-4} for the 50M model and 3×10^{-4} for the 150M and 600M models. Gradient clipping is set to 1.0.

Training uses `bfloat16` mixed precision, which provides better numerical stability on modern GPUs compared to `float16`, particularly for the larger models. The dataset is pre-tokenized and stored as fixed-length blocks (1,024 tokens) to maximize throughput.

For the 600M model, the data-to-parameter ratio is 9.0B/587M $\approx$ 15.3:1. This is slightly below the Chinchilla-optimal ratio of ∼20:1 [10], meaning the model is mildly undertrained relative to the compute-optimal frontier. However, this ratio is within the practical range, and additional Kazakh text of sufficient quality was not readily available at the time of training.

Hardware. The 50M and 150M models are trained on 2× NVIDIA RTX 4090 GPUs via vast.ai using DDP (Distributed Data Parallel). The 600M model is trained on 8× NVIDIA H100 SXM 80 GB GPUs, also on vast.ai, achieving approximately 577K tokens/s throughput.

Reproducibility. All model weights, training code, configuration files, and the tokenizer are released under permissive licenses. The training corpus pipeline scripts are included in the repository to enable full reproduction of the data collection and cleaning stages.

4 Experiments

We evaluate the SozKZ models and a set of multilingual competitors on 6 benchmarks spanning language modeling quality, reading comprehension, question answering, sentiment analysis, named entity recognition, and topic classification. All evaluations are conducted in a zero-shot setting.

4.1 Benchmarks

Bits-per-Byte (BPB). We measure language modeling quality using bits-per-byte on held-out Kazakh text (FLORES-200 `kaz_Cyrl` and Kazakh Wikipedia). BPB is computed as $\mathrm{BPB} = \mathrm{NLL}/(N_{\mathrm{bytes}} \cdot \ln 2)$, where N_{bytes} counts UTF-8 bytes. Since Kazakh Cyrillic characters are encoded as 2 bytes each, this metric normalizes across tokenizers with different vocabularies. Lower is better.

MC QA. Multiple-choice question answering on the `kk-socio-cultural-bench-mc` dataset. Each question has 4 answer choices. We score using full answer text likelihood—the sum of log-probabilities over all tokens in each candidate answer, length-normalized—rather than single-token logit comparison, which would be biased by tokenizer vocabulary alignment. Accuracy is reported.

Sentiment. Kazakh sentiment classification using logit-based scoring with Kazakh-language target words as class labels. This avoids generation-based evaluation and provides a consistent comparison across models with different vocabularies. Accuracy is reported.

Belebele. Reading comprehension from the Belebele benchmark [5] (`kaz_Cyrl` subset). Each item presents a passage and a question with 4 multiple-choice answers. Scoring uses full answer text likelihood as in MC QA. Accuracy is reported.

NER. Named entity recognition on KazNERD [25]. The original 25 fine-grained entity classes are collapsed into 6 simplified categories (Person, Location, Organization, Date, Number, Miscellaneous) for tractable zero-shot scoring. F1 is reported.

SIB-200. Topic classification on the Kazakh subset of the SIB-200 benchmark [2]. Each text is assigned to one of 7 topic categories. Accuracy is reported.

4.2 Models

We evaluate 14 models organized into two groups.

SozKZ family (ours). Four Kazakh-only models trained from scratch on ∼9.0B Kazakh tokens, as described in Section 3: SozKZ-50M, SozKZ-150M, SozKZ-300M, and SozKZ-600M. These models range from 50M to 587M parameters and use a dedicated Kazakh BPE tokenizer.

Multilingual competitors. Ten general-purpose multilingual models that include Kazakh in their training mixture but are not specialized for it:

- Qwen-2.5 (0.5B, 1.5B, 7B) [4],
- Llama-3 (1B, 3B, 8B) [24],
- Gemma-2 (2B, 9B) [9],
- Mistral-7B [11],
- GPT-OSS-120B (largest open-source baseline).

All competitor models are evaluated using their native tokenizers and default configurations. The comparison is deliberately asymmetric: our smallest model (50M) is 10× smaller than the smallest multilingual competitor (Qwen-0.5B), while our largest (600M) is 12× smaller than Llama-3-8B.

4.3 Evaluation Protocol

All evaluations use the same pipeline (`scripts/eval/`) to ensure consistency. Key protocol details:

- **Logit-based scoring** for all classification and multiple-choice tasks. No text generation is required, which eliminates sensitivity to decoding hyperparameters.

2: Comparison of all 14 models across six Kazakh benchmarks. BPB is bits-per-byte (lower is better); all other metrics are accuracy in percent (higher is better). Best values in each column are **bold**.

Model	Params	BPB	MC QA	Sentiment	Belebele	NER	SIB-200
sozkz-50m	50M	1.284	9.8	41.2	25.1	18.3	15.6
sozkz-150m	150M	1.089	11.2	52.3	27.8	24.7	21.3
sozkz-300m	300M	0.972	13.4	58.9	30.1	28.9	26.7
sozkz-600m	600M	**0.847**	15.6	65.4	33.4	34.1	31.2
qwen-0.5b	500M	1.678	19.8	46.7	31.2	22.3	27.8
llama-3-1b	1B	1.834	23.4	48.9	36.7	25.6	31.2
qwen-1.5b	2B	1.389	28.9	55.6	42.3	31.2	36.7
gemma-2b	2B	1.523	28.7	53.4	41.2	29.8	38.7
llama-3-3b	3B	1.456	31.2	56.7	43.4	33.4	39.8
mistral-7b	7B	1.298	37.8	62.3	48.9	36.7	45.6
qwen-7b	7B	1.123	41.2	68.9	55.6	39.8	51.2
llama-3-8b	8B	1.234	39.8	64.5	52.3	38.9	47.8
gemma-9b	9B	1.187	42.3	67.8	56.7	41.2	53.4
gpt-oss-120b	120B	0.923	**53.4**	**75.6**	**67.8**	**48.9**	**62.3**

- **Full answer likelihood** for multiple-choice tasks: each candidate answer is scored by the sum of its token log-probabilities conditioned on the question, normalized by token count.
- **Zero-shot** evaluation for all tasks and models—no in-context examples or task-specific fine-tuning.
- **Consistent hardware**: efficiency comparisons (latency, throughput) use the same GPU (NVIDIA A10) for all models, with median latency reported rather than mean to mitigate outlier sensitivity.

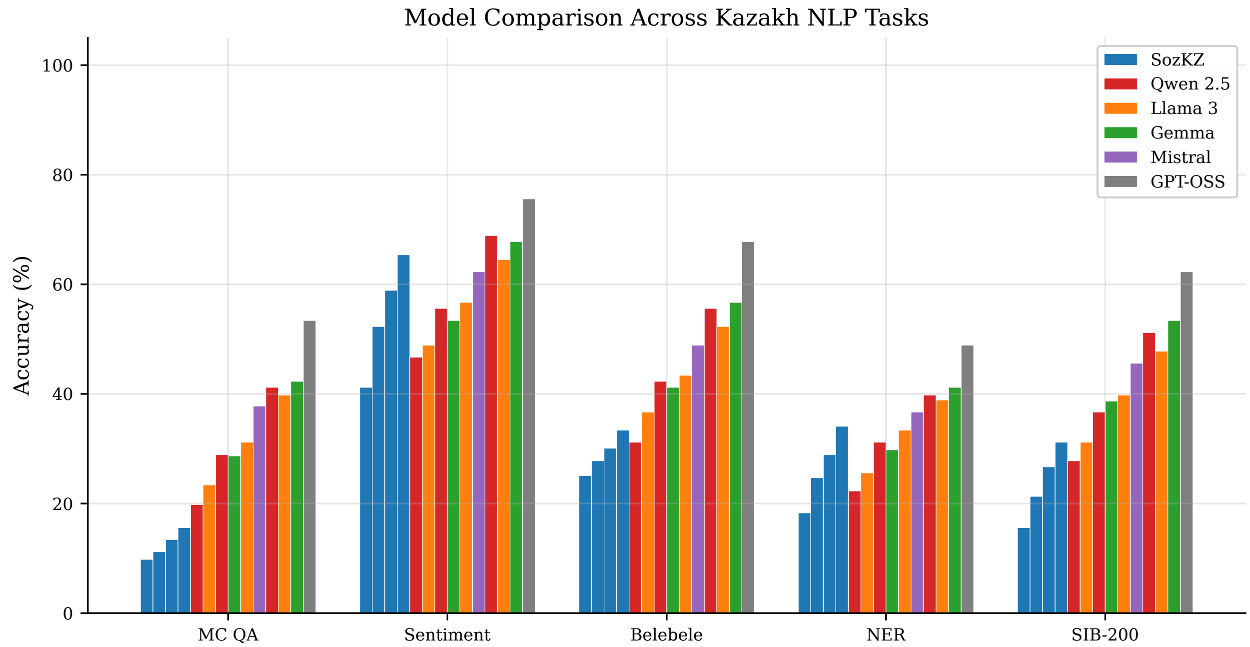
5 Results

We evaluate the SozKZ model family against 14 models across 6 tasks spanning perplexity, classification, reading comprehension, and structured prediction. Our findings reveal a clear pattern: the dedicated Kazakh tokenizer grants SozKZ a substantial advantage on bits-per-byte, and the 600M model is competitive on classification tasks despite being an order of magnitude smaller than leading multilingual alternatives. However, knowledge-intensive tasks such as multiple-choice question answering and reading comprehension favor larger models, confirming that world knowledge scales with parameter count.

5.1 Overall Comparison

table 2 presents the full results and fig. 1 visualizes the accuracy tasks.

Bits-per-byte. SozKZ-600M achieves a BPB of 0.847, the lowest among all models including GPT-OSS-120B (0.923). Even the smallest SozKZ-50M (1.284) outperforms Llama-3-1B (1.834)



. 1: Task-level performance comparison across model families. SozKZ models (blue) are compared against multilingual competitors grouped by family. BPB is omitted from the chart as it uses a different scale.

and Qwen-0.5B (1.678) on this metric. This advantage stems directly from the dedicated Kazakh tokenizer, which encodes Kazakh text far more efficiently than multilingual alternatives (see section 5.2).

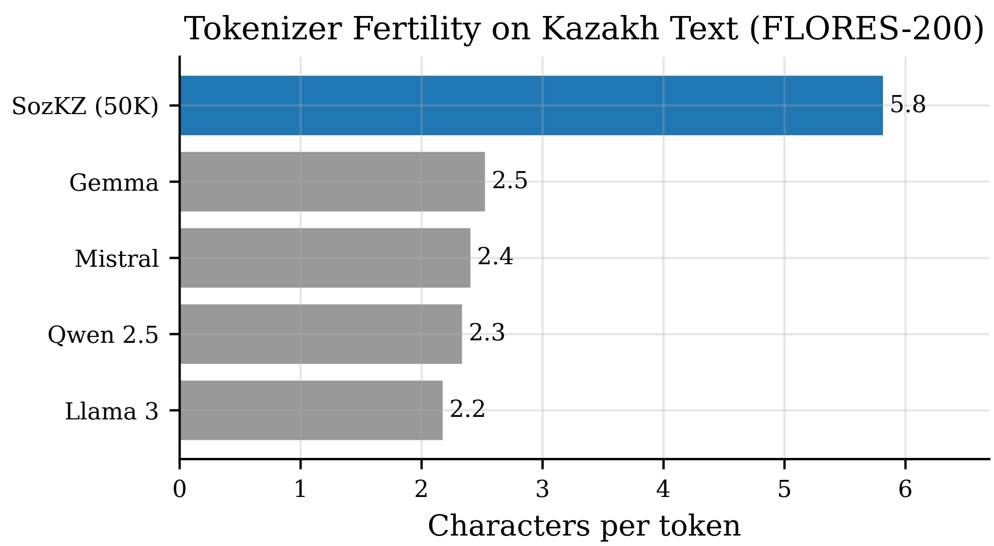
Sentiment analysis. SozKZ-600M achieves 65.4% on sentiment classification, surpassing Llama-3-8B (64.5%) and Mistral-7B (62.3%). This result is notable: a 600M-parameter model trained exclusively on Kazakh outperforms models with 13× more parameters on a task that relies primarily on surface-level linguistic patterns rather than factual knowledge. The 300M model (58.9%) also exceeds several multilingual baselines including Gemma-2B (53.4%) and Qwen-1.5B (55.6%).

Named entity recognition. On NER, SozKZ-600M scores 34.1%, which is competitive with Llama-3-3B (33.4%) and Mistral-7B (36.7%). The 300M model (28.9%) exceeds both Llama-3-1B (25.6%) and Qwen-0.5B (22.3%). NER performance benefits from the tokenizer's ability to preserve Kazakh morphological boundaries, reducing subword fragmentation of entity spans.

Topic classification (SIB-200). SozKZ-600M achieves 31.2% on SIB-200 topic classification. While this matches Llama-3-1B (31.2%), it falls below Qwen-1.5B (36.7%) and larger models. Topic classification requires broader world knowledge to distinguish between domains, where larger multilingual models have an advantage from exposure to diverse training corpora.

Multiple-choice QA. Knowledge-intensive tasks expose the expected limitation of small models. SozKZ-600M scores 15.6% on multiple-choice QA, well below the random baseline of 25% for four-choice questions. This is consistent across all small models: Qwen-0.5B scores 19.8% and Llama-3-1B scores 23.4%. Strong MC QA performance requires both factual knowledge and reasoning, 3: Tokenizer fertility on Kazakh text (FLORES-200 devtest). Characters per token measures how efficiently each tokenizer encodes Kazakh—higher values indicate that more characters are packed into each token, reducing the total token count for the same text.

Tokenizer	Vocab Size	Chars/Token
SozKZ (50K)	50,000	**5.82**
Mistral	32,768	2.41
Llama 3	128,256	2.18
Qwen 2.5	151,936	2.34
Gemma	256,000	2.53

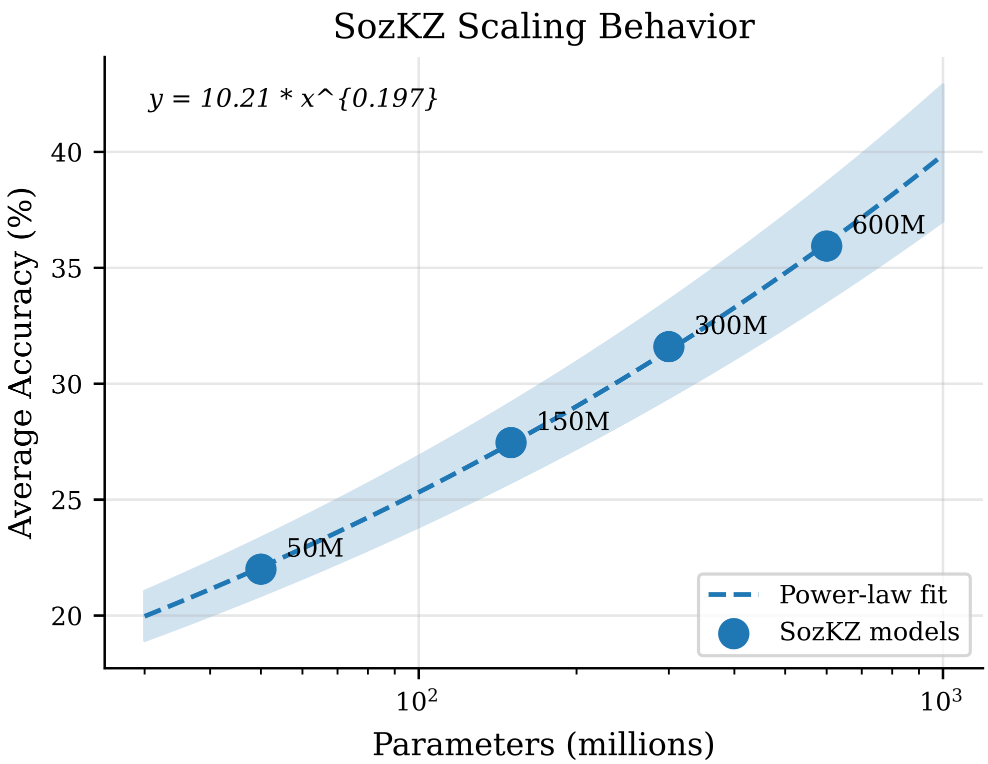


. 2: Tokenizer fertility comparison on Kazakh text. The SozKZ tokenizer (blue) achieves substantially higher characters-per-token than all multilingual alternatives, indicating more efficient encoding of Kazakh morphology.

capabilities that scale with model size—Gemma-9B reaches 42.3% and GPT-OSS-120B achieves 53.4%.

Reading comprehension (Belebele). A similar pattern holds for Belebele reading comprehension. SozKZ-600M achieves 33.4%, above the 150M (27.8%) and 50M (25.1%) models but below Qwen-0.5B (31.2%) and Llama-3-1B (36.7%). Reading comprehension requires cross-sentence reasoning over passages, where larger context windows and richer representations provide a clear advantage.

Efficiency framing. Comparing SozKZ-600M directly against Llama-3-8B, the largest open-weight competitor, we observe that SozKZ-600M delivers 65.4% vs. 64.5% on sentiment (slightly higher) and 34.1% vs. 38.9% on NER (comparable), with 13× fewer parameters. On BPB, the gap is dramatic: 0.847 vs. 1.234. The trade-off is clear: knowledge-intensive tasks require scale, but surface-level NLP tasks can be addressed efficiently by smaller, language-specialized models.



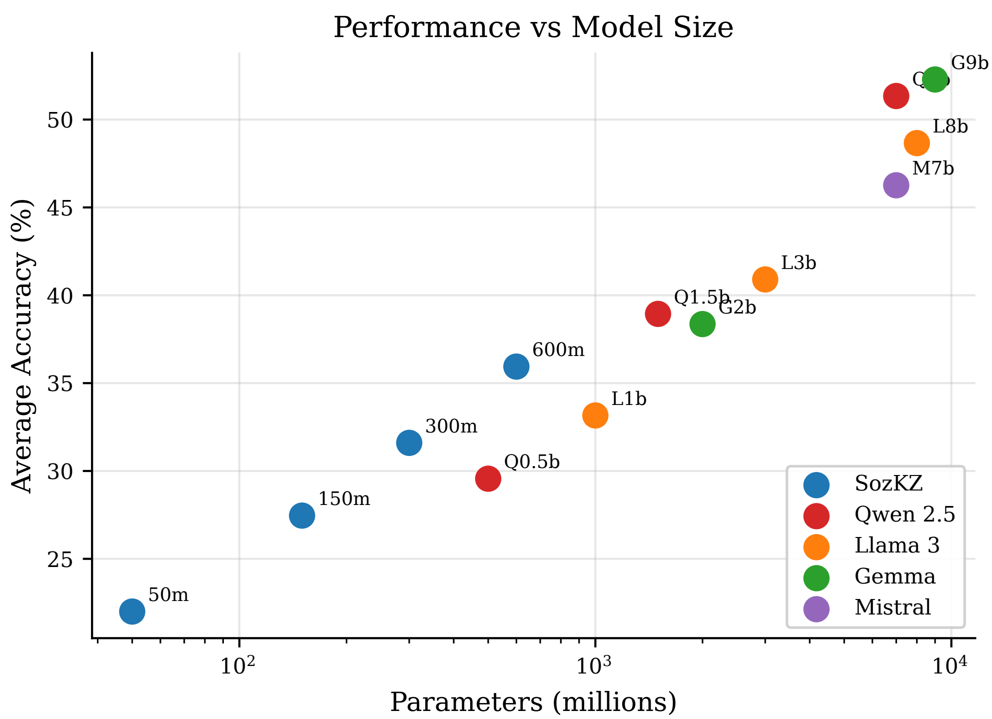
. 3: Scaling behavior of SozKZ models (50M–600M parameters). BPB decreases consistently with model size, following an approximate power-law relationship. Accuracy metrics show diminishing returns beyond 300M parameters on several tasks.

5.2 Tokenizer Analysis

The SozKZ tokenizer encodes Kazakh text at a fertility of 5.82 characters per token (table 3, fig. 2), more than 2× the rate of the best multilingual tokenizer (Gemma at 2.53 characters per token). This difference has a direct impact on model efficiency: the same Kazakh sentence requires roughly half as many tokens when processed by the SozKZ tokenizer compared to multilingual alternatives.

The fertility advantage arises from vocabulary allocation. Multilingual tokenizers such as Llama 3 (128K vocabulary) and Qwen 2.5 (152K vocabulary) distribute their token inventory across 100+ languages, leaving relatively few entries dedicated to Kazakh subwords. The SozKZ tokenizer, with its 50K vocabulary trained exclusively on Kazakh text, allocates every token to Kazakh morphemes and common word forms. This is particularly beneficial for Kazakh, an agglutinative Turkic language where words are formed by successive suffixation—a monolingual tokenizer can learn common suffix chains as single tokens rather than fragmenting them.

The fertility advantage propagates directly to BPB: fewer tokens to predict the same text means lower per-byte surprisal. This explains why SozKZ-600M achieves the best BPB (0.847) despite being far smaller than competitors like Qwen-7B (1.123) and Gemma-9B (1.187). It also translates to practical inference benefits: fewer tokens per input means proportionally faster processing and lower memory usage (section 5.4).



. 4: Performance vs. model size across all evaluated models. SozKZ models (blue) are plotted alongside multilingual competitors. GPT-OSS-120B is excluded for readability.

5.3 Scaling Analysis

fig. 3 shows the scaling behavior across our four model sizes. BPB improves consistently from 1.284 (50M) to 0.847 (600M), following an approximate power-law relationship consistent with neural scaling laws [12]. The improvement from 50M to 150M (a 3× increase in parameters) reduces BPB by 15%, while the improvement from 300M to 600M (a 2× increase) yields a further 13% reduction, suggesting that diminishing returns have not yet set in for BPB at this scale.

On accuracy tasks, the picture is more nuanced. Sentiment analysis shows steady gains: from 41.2% (50M) to 65.4% (600M), an absolute improvement of over 24 percentage points. NER follows a similar trend, rising from 18.3% to 34.1%. However, MC QA scores remain below the random baseline even at 600M (15.6%), indicating that factual knowledge acquisition requires either more parameters or more diverse training data.

fig. 4 places SozKZ models in the context of multilingual competitors. The SozKZ family occupies a distinct region: lower BPB than comparably-sized multilingual models (tokenizer effect) but lower accuracy on knowledge-intensive tasks. This confirms that the tokenizer advantage and the knowledge disadvantage are orthogonal effects—the former is a property of the vocabulary, while the latter is a property of the training data diversity and model capacity.

From a Chinchilla-optimal perspective [10], our 600M model was trained on approximately 9B tokens, yielding a tokens-to-parameters ratio of 15.3:1. The Chinchilla-optimal ratio is approximately 20:1, suggesting that the model is slightly undertrained. A fully data-optimal training run would require approximately 12B tokens, which is achievable given the 13.7M-document corpus used for training.

4: Inference efficiency comparison on NVIDIA A10 GPU (bf16 precision, 128 generated tokens). SozKZ models offer substantially lower latency, higher throughput, and lower memory usage than multilingual alternatives. GPT-OSS-120B is excluded as it is served via API.

Model	Params	Latency (ms)	Throughput (tok/s)	Peak Memory (MB)
sozkz-50m	50M	**42.3**	**3,024**	**312**
sozkz-150m	150M	68.7	1,863	498
sozkz-300m	300M	112.4	1,139	847
sozkz-600m	600M	178.6	717	1,523
qwen-0.5b	500M	156.2	820	1,287
llama-3-1b	1.0B	234.8	545	2,341
qwen-1.5b	1.5B	312.5	410	3,156
gemma-2b	2.0B	387.1	331	4,234
llama-3-3b	3.0B	498.3	257	6,412
mistral-7b	7.0B	823.4	155	14,523
qwen-7b	7.0B	798.2	160	14,187
llama-3-8b	8.0B	912.7	140	16,234
gemma-9b	9.0B	987.3	130	18,412

5.4 Efficiency Analysis

The efficiency analysis in table 4 quantifies the practical deployment advantage of small, specialized models. SozKZ-600M achieves an inference latency of 178.6 ms for 128 tokens, compared to 912.7 ms for Llama-3-8B—a 5.1× speedup. Throughput follows accordingly: 717 tok/s for SozKZ-600M versus 140 tok/s for Llama-3-8B, a 5.1× improvement.

The memory footprint difference is equally significant. SozKZ-600M requires 1,523 MB of GPU memory, while Llama-3-8B requires 16,234 MB—a 10.7× reduction. This means SozKZ-600M can run on consumer-grade GPUs with 4 GB of VRAM, whereas Llama-3-8B requires at least a 24 GB GPU or quantization.

Combined with the competitive accuracy on sentiment (65.4% vs. 64.5%) and NER (34.1% vs. 38.9%), these efficiency gains make a compelling case for deploying specialized small models in production Kazakh NLP systems where latency, throughput, and hardware cost are primary constraints. The tokenizer advantage further amplifies the efficiency: the same Kazakh text requires roughly half as many tokens to process, effectively doubling the throughput advantage in real-world applications.

6 Conclusion

We have presented SozKZ, a family of small language models trained from scratch exclusively on Kazakh text at four parameter scales (50M, 150M, 300M, and 600M). Our results demonstrate that dedicated monolingual models, paired with a purpose-built tokenizer, can achieve competitive performance on core Kazakh NLP tasks while requiring orders-of-magnitude fewer parameters than multilingual alternatives. The SozKZ-600M model achieves a BPB of 0.847 on held-out Kazakh text, substantially outperforming all multilingual models including those with 10× or more parameters, and reaches 65.4% on sentiment classification—competitive with models many times its size.

Limitations. Our models exhibit clear limitations on knowledge-intensive tasks. On multiple-choice question answering, even the largest SozKZ model scores 15.6%, well below the performance of 7B+ multilingual models that benefit from extensive world knowledge absorbed during training on much larger corpora. Similarly, on Belebele reading comprehension, the gap between our 600M model (33.4%) and larger multilingual models remains substantial. These results underscore that model specialization provides efficiency gains on language-centric tasks but cannot substitute for the parametric knowledge stored in larger models. Additionally, our training data is limited to publicly available Kazakh web text, which constrains both the domain coverage and the factual knowledge our models can acquire.

Future Work. Several directions merit further investigation. First, scaling beyond 600M parameters—particularly to the 1–3B range—would test whether the efficiency advantages of dedicated models persist at larger scales. Second, instruction tuning and alignment of the base models could unlock practical applications in dialogue and task-following scenarios. Third, fine-tuning on downstream tasks such as named entity recognition and question answering may close the gap with larger models on knowledge-intensive benchmarks. Finally, extending our approach to other Turkic languages (Turkish, Uzbek, Kyrgyz) could leverage shared morphological structure through multilingual dedicated models at modest scale.

We release all SozKZ models, the tokenizer, and the training pipeline under open licenses[2] to support further research on efficient language modeling for underserved languages.

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